

Unified Roadmap

Consolidated Feature Program

This roadmap tracks the merged target for:

- inventory and local git operations,
- reconciliation and duplicate detection,
- documentation review planning,
- unified operator UX and observability.

Cross-reference:

- [Architecture](#)
- [Migration Plan](#)
- [Features](#)

Milestone Plan

Milestone 1: Platform Baseline (Must)

- Extract and modularize shared scanner/git metadata utilities.
- Preserve reconciliation JSON/CSV/HTML export behavior in new module layout.
- Preserve doc inventory/queue/batch script behavior in new module layout.
- Define canonical API contract for status/reconcile/doc-review/report endpoints.

Dependencies:

- none (foundation).

Milestone 2: Unified API and Compatibility Layer (Must)

- Implement unified API host with route handlers per domain.
- Add compatibility adapters for existing dashboard endpoints.
- Standardize error envelope and correlation ID propagation.
- Implement log + metric emission hooks across all operations.

Dependencies:

- Milestone 1.

Milestone 3: UI Consolidation (Must)

- Integrate Reconciliation and Doc Review domains into dashboard navigation.
- Add run history/artifact panel and domain-level health state indicators.
- Normalize table patterns (filters, grouping, pagination strategy).
- Align empty/error/loading states and operator guidance text.

Dependencies:

- Milestone 2.

Milestone 4: Reliability and Observability Hardening (Should)

- Add health endpoints and dependency health checks.
- Add metrics endpoint and dashboards for run trends/failures.
- Add retry/backoff policy wrappers for remote calls.
- Add schema validation before artifact write.

Dependencies:

- Milestone 2.

Milestone 5: Operationalization (Should)

- Add scheduled task templates for periodic scans and report generation.
- Add retention policies for logs and output artifacts.
- Finalize cutover and deprecation notices for legacy endpoints.

Dependencies:

- Milestones 3 and 4.

Prioritized Backlog (MoSCoW)

Must

- Canonical data contracts for `RepoItem`, `ComparisonItem`, `DocManifestRepo`, `QueueItem`, `BatchPlanItem`.
- Unified config model and secret handling policy.
- Regression coverage for duplicate detection and doc queue scoring.
- UI support for local-only degraded operation when GitHub dependency is unavailable.

Should

- Dual GitHub adapter abstraction (`gh` and API) with fallback logic.
- Incremental scan mode to reduce full-traversal cost.
- Artifact browser enhancements (direct links, retention metadata, size summaries).

Could

- Operator-defined saved views and filters.
- Comparative run analytics (diff between two reconciliation runs).
- Optional notification hooks for failed scheduled runs.

Technical Debt and Refactor Candidates

- Split monolithic reconciliation script into domain modules.

- Remove dashboard backend placeholder endpoints by completing or eliminating them.
- Eliminate duplicate scan logic between dashboard and reconciliation code paths.
- Replace ad-hoc console logging with shared structured logger.
- Normalize output path conventions across all domains.

Observability and Reliability Tasks

- Implement required structured log fields on every operation.
- Emit metric counters/histograms/gauges described in architecture.
- Add liveness/readiness/dependency health checks.
- Introduce error categorization (**validation**, **dependency**, **timeout**, **internal**).
- Add post-retry validation checks and partial-result annotations.

Completion Criteria

- One dashboard and one API endpoint set for all core workflows.
- Script parity retained for key outputs during transition.
- Observability baseline enforced by tests/checks.
- Legacy standalone paths deprecated with documented replacement commands.